Name: _____ Date: _____

PS-02: $\varepsilon - \delta$ Proofs, Part II Solutions

Topics
• $\varepsilon - \delta$ proofs of polynomial functions
• $\varepsilon - \delta$ proofs of rational functions

1. $\lim\limits_{x \to -3} \left(x^2 + 4x + 1\right) = -2$

   Let $\varepsilon > 0$ and choose $\delta = \min\left(1, \frac{\varepsilon}{3}\right)$. Then, if $0 < |x + 3| < \delta$, it must be true that

$$\begin{aligned} |f(x) - L| &= \left|(x^2 + 4x + 1) - (-2)\right| \\ &= \left|x^2 + 4x + 3\right| \\ &= |(x + 3)(x + 1)| \\ &= \underbrace{|x + 3|}_{<\delta} \cdot |x + 1| \end{aligned}$$

   We can find an upper bound on $|x + 3|$ as follows: If

$$|x + 3| < \delta \leq 1 \implies -1 < x + 3 < 1 \implies 1 < x + 1 < 3 \implies |x + 1| < 3$$

   Then,

$$|f(x) - L| = \ldots = |x + 3| \cdot |x + 1| < \delta \cdot 3 \leq \frac{\varepsilon}{3} \cdot 3 = \varepsilon \quad \square$$

2. $\lim\limits_{x \to 2} \left(x^3 - 7x\right) = -6$

   Let $\varepsilon > 0$ and choose $\delta = \min\left(1, \frac{\varepsilon}{12}\right)$. Then, if $0 < |x - 2| < \delta$, it must be true that

$$\begin{aligned} |f(x) - L| &= \left|(x^3 - 7x) - (-6)\right| \\ &= \left|x^3 - 7x + 6\right| \\ &= |(x - 2)(x + 3)(x - 1)| \\ &= \underbrace{|x - 2|}_{<\delta} \cdot |x + 3| \cdot |x - 1| \end{aligned}$$

   We can find an upper bound on the other two factors as follows:

$$|x - 2| < \delta \leq \implies -1 < x - 2 < 1 \implies 4 < x + 3 < 6 \implies |x + 3| < 6$$
$$|x - 2| < \delta \leq \implies -1 < x - 2 < 1 \implies 0 < x - 1 < 2 \implies |x - 1| < 2$$

   You can confirm the upper bounds with this Desmos calculator:
   `https://www.desmos.com/calculator/zyaxx5fihr`.

   Without Desmos, note that the function $g(x) = (x + 3)(x - 1)$ is continuous, and so it attains a maximum on the closed interval $[1, 3]$ (which is the interval of x values that satisfy $|x - 2| < 1$). The maximum occurs at one of the endpoints, and so we can check $g(1) = 4$ and $g(3) = 12$, which confirms our upper bound of 12.

Then,

$$|f(x) - L| = \ldots = |x-2| \cdot |x+3| \cdot |x-1| < \delta \cdot 6 \cdot 2 = 12\,\delta = 12 \cdot \frac{\varepsilon}{12} = \varepsilon \quad \square$$

3. $\lim\limits_{x \to 4} \left(\dfrac{x^2 - 16}{x - 4} \right) = 8$

Let $\varepsilon > 0$ and choose $\delta = \varepsilon$. Then, if $0 < |x - 4| < \delta$, it follows that

$$\begin{aligned} |f(x) - L| &= \left| \frac{x^2 - 16}{x - 4} - 8 \right| \\ &= \left| (x+4) \cdot \underbrace{\frac{x-4}{x-4}}_{=1 \text{ since } |x-4|>0} - 8 \right| \\ &= |(x+4) - 8| \\ &= |x - 4| \\ &< \delta \\ &= \varepsilon \quad \square \end{aligned}$$

4. $\lim\limits_{x \to -1} \left(\dfrac{x^2 - 7x + 4}{x - 3} \right)$

Let $\varepsilon > 0$. Choose $\delta = \min\left\{1, \dfrac{3}{7}\varepsilon\right\}$. Then, if $0 < |x+1| < \delta$, we have

$$\begin{aligned} |f(x) - L| &= \left| \frac{x^2 - 7x + 4}{x - 3} + 3 \right| \\ &= \left| \frac{x^2 - 7x + 4}{x - 3} + \frac{3x - 9}{x - 3} \right| \\ &= \left| \frac{x^2 - 4x - 5}{x - 3} \right| \\ &= \left| \frac{(x-5)(x+1)}{x - 3} \right| \\ &= \underbrace{|x+1|}_{<\delta} \cdot \left| \frac{x-5}{x-3} \right| \end{aligned}$$

Thus, we'll need to bound the second factor, $\left| \dfrac{x-5}{x-3} \right| = |x-5| \cdot \dfrac{1}{|x-3|}$.

(a) $|x+1| < 1 \implies -1 < x+1 < 1 \implies -7 < x-5 < -4 \implies |x-5| < 7$

(b) $|x+1| < 1 \implies -1 < x+1 < 1 \implies -5 < x-3 < -3 \implies -\dfrac{1}{3} < \dfrac{1}{x-3} < -\dfrac{1}{5} \implies \left| \dfrac{1}{x-3} \right| < \dfrac{1}{3}$

and we have

$$\left| \frac{x-5}{x-3} \right| = |x-5| \cdot \frac{1}{|x-3|} < 7 \cdot \frac{1}{3} = \frac{7}{3}$$

so that

$$|f(x) - L| = \ldots = |x+1| \cdot \left| \frac{x-5}{x-3} \right| < \delta \cdot \frac{7}{3} \leq \frac{3}{7}\varepsilon \cdot \frac{7}{3} = \varepsilon \quad \square$$

Here's a different way to found the expression $\left|\dfrac{x-5}{x-3}\right|$ on the interval $(-2,0)$:

- $x-5<0$ on $(-2,0)$
- $x-3<0$ on $(-2,0)$

Thus, $\left|\dfrac{x-5}{x-3}\right| = \dfrac{x-5}{x-3}$ on $(-2,0)$. Let $g(x) = \dfrac{x-5}{x-3} = 1 - \dfrac{2}{x-3}$ on $(-2,0)$ and check its monotonicity:

$$g'(x) = \frac{d}{dx}\left(\frac{-2}{x-3}\right) = \frac{2}{(x-3)^2} > 0 \text{ for all } x \in (-2,0)$$

Therefore $g(x)$ is strictly inreasing on $(-2,0)$ and so the maximum occurs at $x=0$. Evaluating $g(0) = \dfrac{0-5}{0-3} = \dfrac{5}{3}$, we have $\left|\dfrac{x-5}{x-3}\right| \leq \dfrac{5}{3}$ on $(-2,0)$.

Here's a Desmos Calculator